Answer **one** question in this section in the spaces provided.

7 **(a)** In a continuous-flow experiment to measure the specific heat capacity of water, water flows through a tube containing an electrical heater.

Water flows at a rate of 0.150 kg min^{-1} and is heated at a rate of 25.2 W. The steady temperatures of the water entering and leaving are 15.2 °C and 17.4 °C respectively.

The rate of flow is then increased to 0.232 kg min^{-1} and the rate of heating to 37.8 W. The steady temperatures of the water entering and leaving the tube remain unchanged.

(i) State what is meant by *specific heat capacity*.

.....

.....

..... [1]

(ii) This experiment was repeated under different flow rate and heater power.

Explain why it is essential for the temperatures of water entering and leaving to remain unchanged.

.....

.....

..... [1]

(iii) Hence, determine the specific heat capacity of water.

specific heat capacity of water = ... J kg^{-1} K^{-1} [3]

(b) A volume of ideal gas at 30.0 °C is enclosed in a metal cylinder fitted with a 8.00 kg freely moving piston of cross sectional area 60.0 cm^2 as shown in Fig. 7.1a. Atmospheric pressure is 100 kPa.

When the gas is slowly heated from 30.0 °C to 100.0 °C, the piston rises 20.0 cm as shown in Fig. 7.1b.

The piston is then fastened in place, and the gas is allowed to cool back to 30.0 °C as shown in Fig. 7.1c.

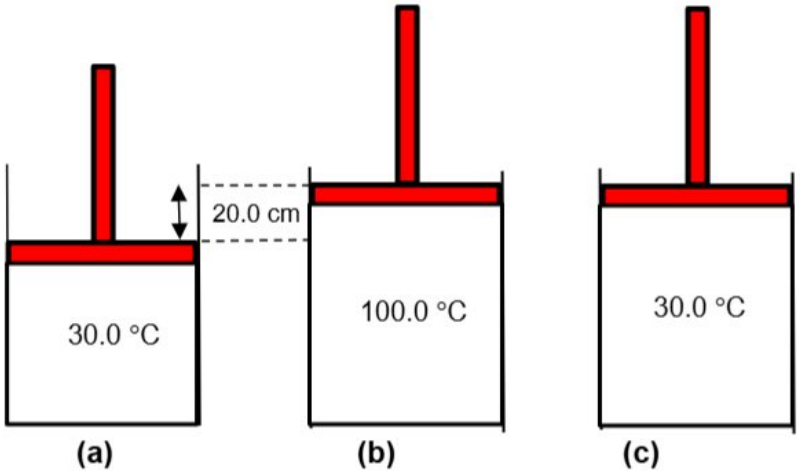


Fig. 7.1

(i) Describe the thermodynamic process that represent the state changes from

1. Fig. 7.1(a) to Fig. 7.1(b)

..... [1]

2. Fig. 7.1(b) to Fig. 7.1(c)

..... [1]

(ii) Calculate the pressure p of the gas shown in Fig. 7.1(b) before the gas was cooled.

p = ... Pa [3]

(iii) Sketch the variation with volume of the pressure of the gas for the processes described in **(b)(i)** in Fig. 7.2**.** Label them with X and Y respectively for the first and second process.



Fig. 7.2

[2]

(iv) Using the first law of thermodynamics, determine the net heat gained by the gas for the processes shown in Fig. 7.1a to Fig. 7.1c

net heat gained = ... J [4]

(c) Fig. 7.3 shows a model rocket for demonstrating the principle of rocket propulsion. Air is pumped into an upside-down plastic bottle that has been filled with water. When the pressure reaches 3.6×10^5 Pa, the air valve is forced out by the water pressure and the air in the bottle expands.

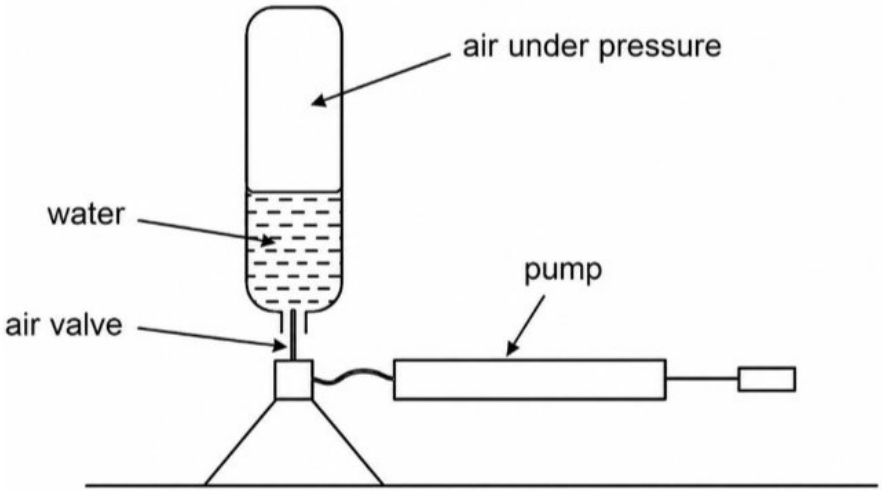


Fig. 7.3

(i) Explain why the force on the rocket due to the expelled water decreases as the air expands.

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.....

..... [2]

(ii) Fig. 7.4 shows the variation of pressure with volume for the air initially in the bottle as it expands from 3.6×10^5 Pa to atmospheric pressure.

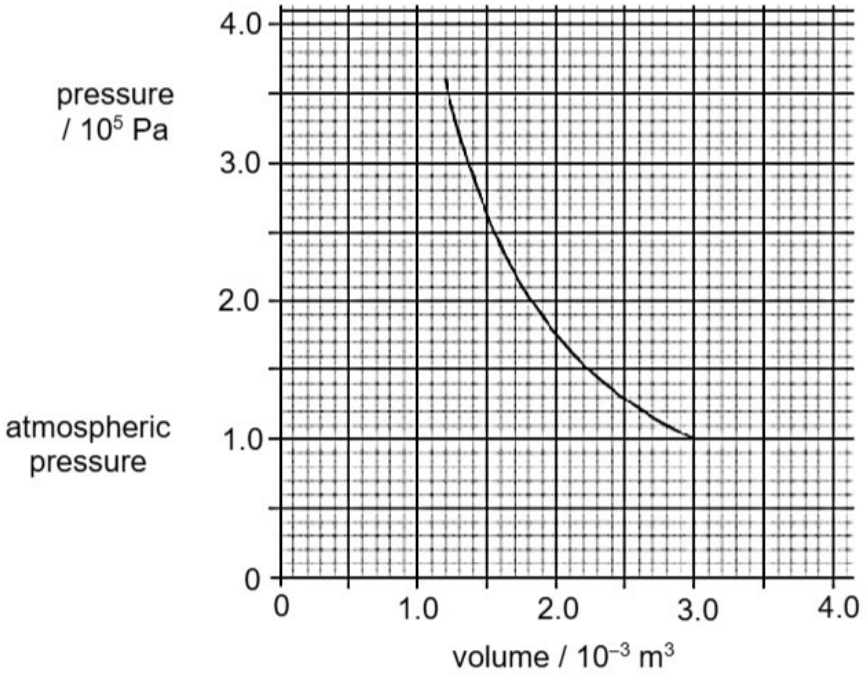


Fig. 7.4

With reference to Fig. 7.4, describe how the temperature of the air varies during its expansion. Support your answer with appropriate calculations.

.....

.....

..... [2]